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Ex No: 8	Real-Time Data Processing in Snowflake
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Implementing Real-Time Data Processing in Snowflake Using Streams and Tasks

Objective:

This lab experiment aims to demonstrate how real-time data processing can be implemented in Snowflake using Streams and Tasks. The goal is to build an automated data-cleaning workflow that continuously monitors, cleans, and standardizes hospital patient records. It helps ensure that data remains consistent, accurate, and up to date without manual intervention.

Outcomes:

- Understand how Streams in Snowflake track real-time data changes (INSERT, UPDATE, DELETE).
- Learn how Tasks are used to schedule and automate data processing workflows.
- Design and implement an end-to-end real-time data cleaning process for hospital records.
- Demonstrate automation in a healthcare context using Snowflake's stream-task architecture.

Materials:

- Snowflake environment or simulation setup.
- Sample hospital dataset with inconsistent values and formats.
- Access to SQL worksheet for executing stream and task operations.
- Documentation for TRY_TO_NUMBER, TRY_TO_DATE, and CASE functions in Snowflake.

Lab Procedure:

Stage 1: Problem Definition and Requirements Gathering

The hospital wants to maintain standardized patient records across various data sources. The business goal is to automate the process of cleaning messy or inconsistent data—such as incorrect dates, invalid bill amounts, or text-based ages so that all records are reliable for downstream analytics. The analyst defines the data attributes (patient_id, name, age, gender, visit_date, diagnosis_code, bill_amount) and the cleaning requirements.

Stage 2: Data Model Design

The engineer designs the database structure in Snowflake. A database named HOSPITAL_DB and a schema named PATIENT_SCHEMA are created to organize the data. Two tables are defined: RAW_PATIENTS for unprocessed data (storing all fields as strings) and CLEAN_PATIENTS for cleaned data (with correct data types and an additional processed_at column). The separation ensures traceability and avoids overwriting raw data.

Stage 3: Stream Setup

A Stream is created on the RAW_PATIENTS table to capture new inserts or updates. This stream acts as a change tracker, recording data modifications since the last processing cycle. It ensures that only new or changed records are considered for cleaning, reducing redundant computations and improving efficiency.

Stage 4: Data Cleaning and Transformation

A Snowflake Task is designed to read new records from the stream and perform automated cleaning. Transformation rules include converting age values from text to integers, removing commas from bill_amount fields, standardizing date formats (e.g., from DD-MM-YYYY or MM/DD/YYYY to YYYY-MM-DD), and replacing invalid or missing entries with defaults. The cleaned records are then inserted into the CLEAN_PATIENTS table along with a timestamp.

Stage 5: Task Scheduling and Automation

The cleaning Task is scheduled to run every 10 minutes. This ensures that newly inserted data is processed automatically without manual execution. Such automation minimizes human effort and maintains near real-time synchronization between raw and cleaned datasets.

Stage 6: Data Insertion and Validation

Sample messy data entries are inserted into RAW_PATIENTS to test the workflow. Examples include missing values, text-based ages like 'Twenty-Five', or invalid bill amounts such as 'abc'. After the Task runs, the CLEAN_PATIENTS table is checked to

confirm that all fields are standardized and cleaned. The analyst validates accuracy by comparing before-and-after transformations.

Stage 7: Verification and Monitoring

The team retrieves all records from CLEAN_PATIENTS ordered by patient_id to confirm the cleaning logic worked correctly. They verify that numeric fields are properly formatted, dates are standardized, and null or missing entries have been appropriately handled. Continuous monitoring ensures that future data changes trigger the same automated pipeline.

Reflection:

The Snowflake stream-task architecture ensures real-time, consistent, and accurate data processing by integrating continuous change tracking with automated task execution. Streams detect every data modification on a source table, enabling downstream tasks to respond instantly. Tasks then perform scheduled or event-driven cleaning, transformation, or loading operations.

In the hospital scenario, this means that every new patient record, whether added or updated, is automatically captured and cleaned within minutes. The architecture guarantees that healthcare data remains accurate and analysis-ready without manual intervention—critical for operational and clinical decisions.

Compared to traditional batch ETL, which processes data at fixed intervals, the stream-task model delivers near-real-time updates. Batch ETL often leads to delayed insights and potential inconsistencies between data loads. In contrast, the Snowflake approach maintains continuous synchronization between raw inputs and cleaned datasets, offering a highly reliable and scalable foundation for healthcare data management.

GitHub Link: